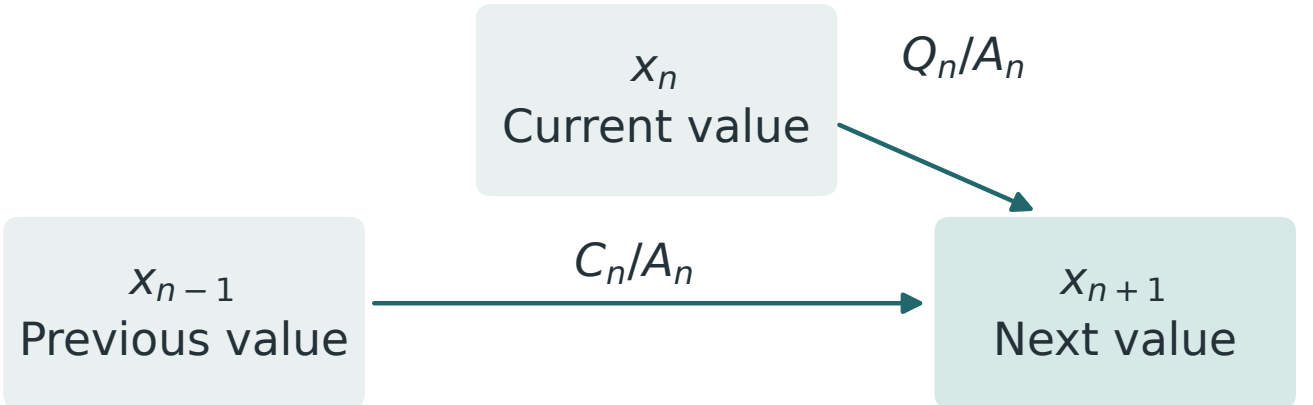

$$A_n x_{n+1} = Q_n x_n + C_n x_{n-1}$$

Both u and v obey the same recurrence, from distinct initial data.